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Roll No. ....

6th Sem/
ECE, ECE (For Speech & Hearing Impaired)
Subject : Embedded Systems

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which memory storage is widely used in PCs and Embedded Systems?

a) EEPROM b) Flash memory

c) SRAM d) DRAM

Q.2 LCD stands for

a) Light coded display

b) Liquid crystal display

c) Liquid converted data

d) Light carrying data

Q.3 Which of the following should a micro controller at-leaset should consist of?

a) CPU, ROM, I/O ports and timers

b) RAM, ROM, I/O ports and timers

c) CPU, RAM, I/O ports and timers d)

Q.4 Which of the following is not a type of memory?

a) RAM b) FPROM

c) EEPROM d) ROM

Q.5 When the micro controller executes some arithmetic operations, then the flag bits of which register are affected?

a) PSW b) SP

c) DPTR d) PC

Q.6 Pic 18F458 has ____ no's of I/O pins.

a) 40 b) 35

c) 33 d) 27

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write full form of ROM.

Q.8 What is purpose of SFRs register?

Q.9 Define Embedded System.

Q.10 Write any two names of Advanced Micro controllers.

Q.11 PIC is _____ Pin IC.

Q.12 Write full form of PIC.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain data serialization in C.

Q.14 Explain the architecture of a typical PIC micro controller.

Q.15 Write any four arithmetic instruction, explain any one instruction.

Q.16 What is difference between Serial and Parallel Data input/output.

Q.17 Write any five advantages of an Embedded system.

Q.18 Write short note on History and features of PIC micro controllers.

Q.19 What are the key features and advantages of programming PIC micro controllers using C?

Q.20 Explain any four data type in C.

Q.21 Discuss the challenges and considerations when interfacing a PIC18 micro controller with a DC motor.

Q.22 Describe the process of inter facing a PIC18 micro controller with DAC and sensor.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Draw pin diagram of PIC 18F458 and explain any 3 pins.

Q.24 Draw embedded system architecture and write any four functions of embedded system.

Q.25 Write short note on :

a) Relays b) Opto isolator